An AI Pipeline for Automated Assessment of Lumbar Spinal Stenosis Using Multi-Institutional MRI

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***Abstract*— Lumbar spinal stenosis requires accurate localization and severity assessment for effective clinical decision-making. We propose an AI pipeline for automated stenotic region localization and severity classification using axial and sagittal MRI images. The system explicitly accounts for anatomical differences between imaging planes by applying plane-specific processing for stenotic region detection and subsequent severity grading. The detected regions of interest are subsequently classified into four severity categories. Experimental results using multi-institutional MRI data demonstrated robust stenotic region localization and stable severity classification performance for both central canal and nerve root canal stenosis.**

Keywords—degenerative lumbar spinal stenosis, spine MRI, automated diagnosis, deep learning, clinical decision support

I. INTRODUCTION

The global population is undergoing a rapid demographic shift toward aging due to declining birth rates and increasing life expectancy [1]. As a result, healthcare systems worldwide face a growing burden in managing age-related spinal disorders and degenerative changes commonly observed in elderly populations [2]. Among these conditions, degenerative lumbar spinal stenosis is a major cause of pain and reduced quality of life in older adults [3], [4]. Accurate identification of stenotic locations and differential diagnosis are essential for determining appropriate treatment strategies for this condition [5].

Magnetic resonance imaging (MRI) is the most widely used radiological modality for evaluating lumbar spinal stenosis [6], [7]. However, current clinical practice relies heavily on subjective visual interpretation by experienced specialists, which inevitably leads to inter-observer variability and inconsistency in diagnostic assessments [8]. In addition, the need to comprehensively evaluate multiple imaging sequences and planes substantially increases interpretation time, thereby imposing a significant workload burden on routine clinical practice [9]. Collectively, these factors hinder the establishment of objective and standardized diagnostic criteria.

Recent advances in artificial intelligence (AI)-based medical image analysis have attracted attention as a promising approach to address these limitations [10]. AI models trained on large-scale imaging datasets can automate lesion detection and classification tasks, potentially improving the consistency and reproducibility of spine MRI interpretation [11]. While several prior studies have applied deep learning–based detection or classification models to spine MRI for identifying disc pathologies or stenotic regions, these approaches are typically designed as isolated tasks and do not provide an integrated framework that links stenotic region localization with clinically meaningful severity grading. Moreover, many existing studies are limited to single-institution datasets, rely on a single imaging plane, or focus on coarse binary decision making, which constrains their robustness and clinical applicability [11],[12].

In this study, we propose an automated detection–classification integrated AI pipeline based on T2-weighted MRI for lumbar spinal stenosis assessment. The proposed framework automatically detects stenotic regions in the central canal and nerve root canal and subsequently classifies the severity of stenosis into four discrete grades. By enabling automated diagnosis and severity grading, this approach aims to reduce the interpretation burden on clinicians and promote greater standardization in the diagnostic workflow.

II. MATERIALS AND METHODS

A. Dataset

The dataset used in this study consisted of a multi-institutional collection of lumbar spine MRI scans, including both sagittal and axial planes. MRI data were collected from 2,480 patients who underwent spine MRI examinations at five hospitals between January 2009 and December 2020 and demonstrated radiological findings consistent with degenerative lumbar spinal stenosis. The study protocol was approved by the Institutional Review Board of the Catholic University of Korea (IRB No. XC21WNDE0083), and the requirement for informed consent was waived due to the retrospective nature of the study. All procedures were conducted in accordance with the relevant guidelines and regulations of the Declaration of Helsinki. All MRI data were acquired and stored in the Digital Imaging and Communications in Medicine (DICOM) format.

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B. Overall system architecture

Rather than treating stenosis detection and severity grading as independent tasks, the proposed system is designed as a structured, view-aware pipeline that explicitly incorporates anatomical characteristics associated with different MRI planes. The overall workflow consists of sequential but functionally decoupled stages for view classification, stenotic region localization, and severity grading.

As shown in Fig. 1, input MRI scans are first automatically classified into axial and sagittal views. This view classification step enables the system to route each image to anatomically specialized detection models, reflecting the fact that clinically relevant stenotic regions differ by imaging plane [13].

The bounding boxes predicted during the detection stage are used to generate regions of interest (ROIs) for subsequent severity assessment, while simultaneously serving as a means to initially identify the locations of stenosis. In this process, to minimize variability caused by differences in imaging acquisition conditions across multi-institutional data, the field-of-view is uniformly normalized in millimeter units using pixel spacing information.

The cropped ROI images are then used as inputs to a classification model, in which the severity of central canal stenosis is evaluated for axial images and the severity of nerve root canal stenosis is evaluated for sagittal images. All classification results are reported in four categories: Normal, Mild, Moderate, and Severe.

In this manner, the proposed system is designed to perform automated diagnosis and severity assessment of lumbar spinal stenosis in a more structured and consistent way, and to improve the efficiency of the clinical diagnostic process.

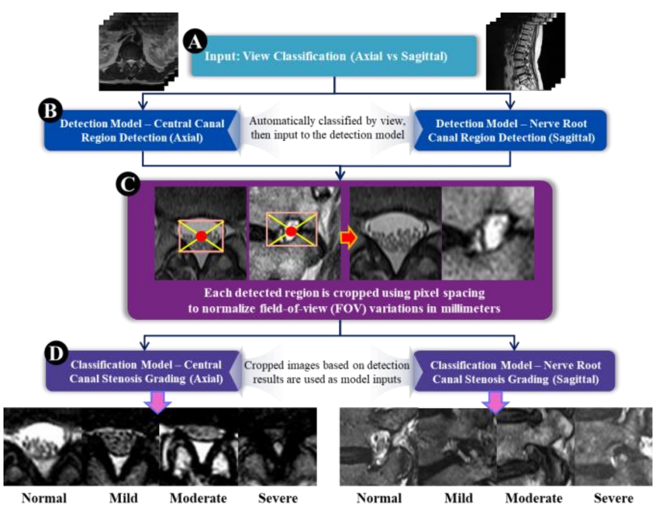

Fig. 1. Overall workflow of the proposed detection–classification pipeline for lumbar spinal stenosis

C. Stenotic Region Localization

In this study, stenotic region localization was formulated as an object detection task to precisely identify candidate regions associated with lumbar spinal stenosis prior to severity grading. To reflect anatomical differences across MRI planes, view-specific detection models were independently designed and trained for axial and sagittal images.

For axial MRI images, a YOLOv11-based detection model was employed to localize stenotic regions within the central canal. For sagittal MRI images, a separate YOLOv11-based detection model was used to identify stenosis within the nerve root canal. This plane-specific modeling strategy enables each detector to focus on distinct anatomical structures and spatial patterns relevant to stenosis localization.

As illustrated in Fig. 2, the YOLOv11 architecture combines hierarchical feature extraction with multi-scale feature aggregation, allowing robust localization of stenotic regions under varying anatomical appearances and imaging conditions [14], [15]. The output of this stage consists of bounding boxes that delineate potential stenotic regions, which are subsequently used for region-of-interest generation in the severity classification stage.

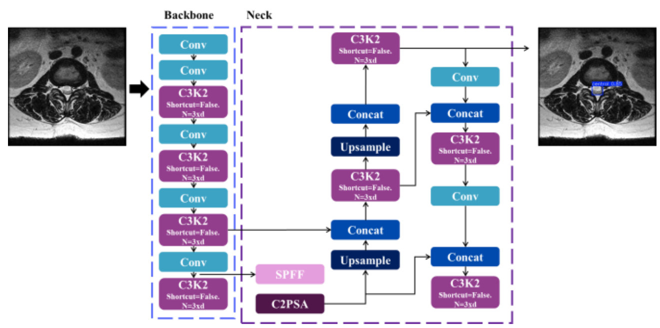

Fig. 2. YOLOv11-based architecture for stenotic region localization in lumbar spine MRI.

D. Stenosis Severity Classification

Following stenotic region localization, a stenosis severity classification stage was implemented to quantitatively assess the degree of stenosis within the localized regions. To ensure stable learning of severity-related features, the classification model was trained using region-of-interest (ROI) images cropped based on ground-truth annotations. This ground-truth–based cropping strategy was adopted during training to eliminate the influence of localization errors and to focus the learning process on discriminative severity patterns.

During inference within the proposed detection–classification pipeline, ROI images were generated by cropping MRI scans according to the bounding boxes predicted by the stenotic region localization models. The classification model trained on ground-truth–based crops was then applied to these detection-based ROIs, reflecting a realistic clinical deployment scenario in which manual annotations are not available.

For axial MRI images, the model assessed the severity of central canal stenosis, whereas for sagittal MRI images, the severity of nerve root canal stenosis was evaluated. In both cases, stenosis severity was categorized into four grades: Normal, Mild, Moderate, and Severe. As illustrated in Fig. 3, a DINOv2-based classification framework with a Vision Transformer backbone was employed to effectively capture morphological variations associated with different levels of stenosis severity [16], [17].

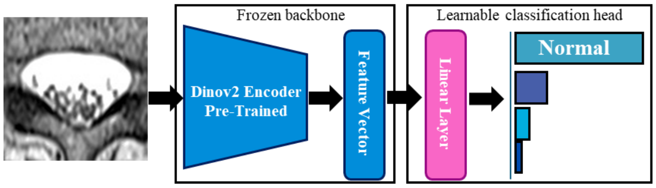

Fig. 3. DINOv2-based architecture for stenosis severity classification.

E. Model Training and Implementation Details

All experiments were conducted in a Python-based environment using PyTorch (version 2.5.1) with CUDA (version 12.8) and cuDNN (version 8.6.0). Deep learning model training was performed using an NVIDIA A100 GPU with 80 GB of memory (NVIDIA, Santa Clara, CA, USA). For image processing and visualization of model performance, OpenCV (version 4.1.1), pydicom (version 2.1.2), scikit-learn (version 0.23.2), and Matplotlib (version 3.3.2) were utilized.

The dataset was split at the patient level into training, validation, and test sets. The validation set was used for early stopping and model selection to prevent overfitting.

For stenotic region localization, two separate YOLOv11-based detection models were trained for central canal localization on axial images and nerve root canal localization on sagittal images. Both models shared a common input resolution of 640×640, an SGD optimizer, and an initial learning rate of 2×10^{-4}. The batch size was set to 64 for the central canal model and 8 for the nerve root canal model. The maximum number of training epochs was set to 500 for both models.

Early stopping was applied with a patience of 100 epochs based on validation performance to prevent overfitting. As a result, the central canal detection model converged after 101 epochs, requiring approximately 2.599 hours of training, while the nerve root canal detection model converged after 103 epochs with a training time of approximately 1.730 hours.

For stenosis severity classification, a DINOv2-based classification model with a Vision Transformer (ViT) backbone was employed for both axial and sagittal views. The model was trained using region-cropped MRI inputs resized to 224×224, with a learning rate of 1×10^{-5} and a batch size of 16.

The maximum number of training epochs was set to 500, and early stopping was applied with a patience of 70 epochs based on validation performance. As a result, the central canal classification model converged at epoch 82, while the nerve root canal classification model converged at epoch 101.

III. RESULTS

The stenotic region localization performance was evaluated using Precision, Recall, and mAP50, as summarized in Table I for both central canal and nerve root canal models.

TABLE I. DETECTION PERFORMANCE OF THE YOLOV11-BASED MODELS FOR STENOTIC REGION LOCALIZATION IN CENTRAL CANAL AND NERVE ROOT CANAL

Location	Precision	Recall	mAP50
Central Canal	0.994	0.990	0.995
Nerve Root Canal	0.751	0.862	0.833

The central canal detection model achieved excellent performance, with a Precision of 0.994, Recall of 0.990, and mAP50 of 0.995, which can be attributed to the clear anatomical definition and consistent morphological characteristics of the central canal on axial MRI images.

In contrast, the nerve root canal detection model showed comparatively lower performance (Precision 0.751, Recall 0.862, mAP50 0.833), likely due to its narrow anatomical structure and ambiguous boundaries with surrounding soft tissues and osseous components. Despite this challenge, the model maintained a relatively high recall, indicating effective identification of candidate stenotic regions.

(a)

(b)

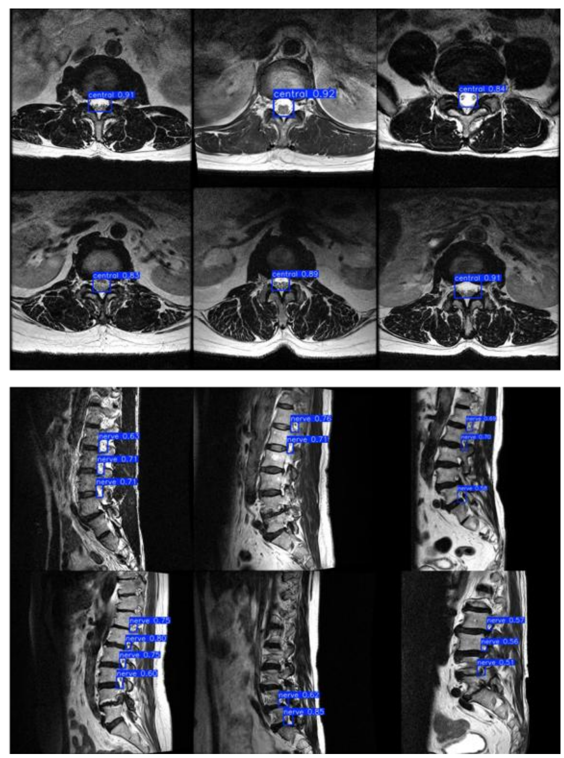

Fig. 4. Qualitative examples of stenotic region localization using YOLOv11. (a) Central canal detection on axial MRI, (b) Nerve root canal detection on sagittal MRI. Blue bounding boxes indicate the localized stenotic regions, and the overlaid values represent the corresponding detection confidence scores.

Fig. 4 presents representative qualitative examples of YOLOv11-based stenotic region localization, demonstrating accurate detection of central canal stenosis in axial MRI and nerve root canal stenosis in sagittal MRI.

Table II summarizes the performance of stenosis severity classification for central canal and nerve root canal MRI images. The proposed DINOv2-based classification models demonstrated stable and consistent performance across both anatomical regions, with all evaluation metrics reported alongside 95% confidence intervals.

TABLE II. CLASSIFICATION PERFORMANCE OF STENOSIS SEVERITY GRADING FOR CENTRAL CANAL AND NERVE ROOT CANAL MRI IMAGES, REPORTED WITH 95% CONFIDENCE INTERVALS.

	Central Canal	Nerve Root Canal
Accuracy (95% CI)	0.8920 (CI: 0.8376 – 0.9298)	0.9034 (CI: 0.8508 – 0.9388)
Sensitivity (95% CI)	0.9002 (CI: 0.8441 – 0.9343)	0.8468 (CI: 0.7860 – 0.8924)
Precision (95% CI)	0.8782 (CI: 0.8180 – 0.9160)	0.9057 (CI: 0.8508 – 0.9388)
F1-score (95% CI)	0.8877 (CI: 0.8310 – 0.9252)	0.8673 (CI: 0.8051 – 0.9066)

For central canal stenosis, the model achieved an accuracy of 0.8920 (95% CI: 0.8376–0.9298) and an F1-score of 0.8877 (95% CI: 0.8310–0.9252), indicating reliable discrimination across the four severity grades. Sensitivity and precision were well balanced, suggesting robust detection of clinically relevant stenotic changes.

For nerve root canal stenosis, the classification performance remained comparable, with an accuracy of 0.9034 (95% CI: 0.8508–0.9388) and an F1-score of 0.8673 (95% CI: 0.8051–0.9066). Although sensitivity was slightly lower than that of the central canal model, precision was higher, reflecting the model's conservative but accurate severity prediction in anatomically constrained regions.

Overall, these results indicate that the proposed severity grading framework maintains consistent classification performance across different anatomical targets and imaging planes.

Fig. 5 shows representative attention maps from the DINOv2-based severity classification model. The model primarily attends to anatomically relevant regions, focusing on the central canal in axial images and the nerve root canal in sagittal images, indicating that stenosis severity is assessed based on clinically meaningful structures.

(a)

(b)

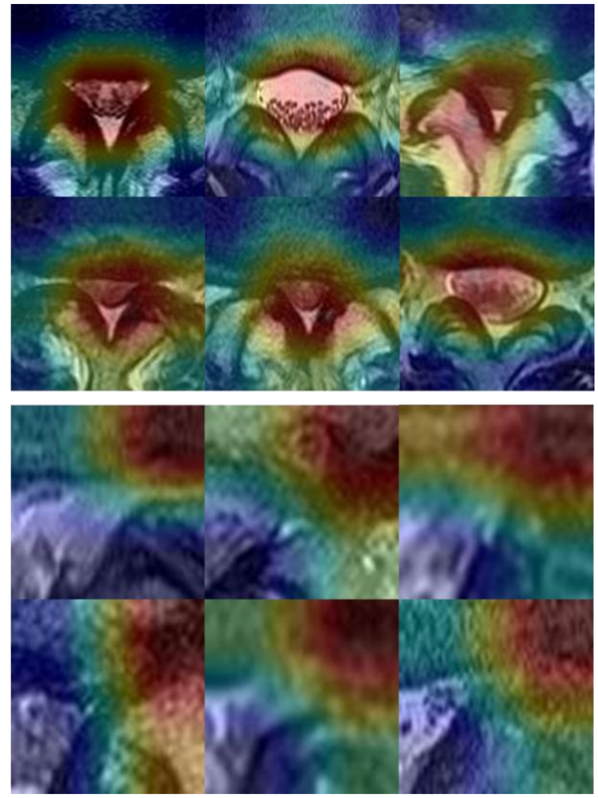

Fig. 5. Attention map visualization of the DINOv2-based stenosis severity classification model. (a) Central canal, (b) Nerve root canal.

Fig. 6 shows the final output of the proposed AI pipeline for a single patient.

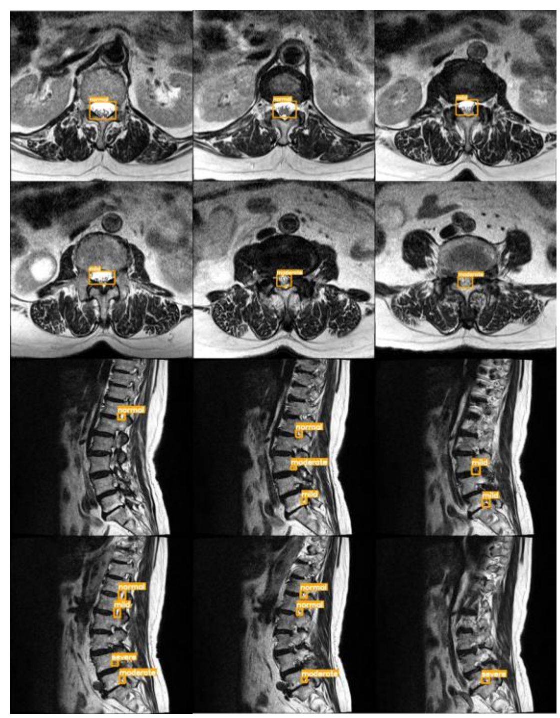

Fig. 6. Representative example of the proposed AI pipeline output for a single patient. Orange bounding boxes indicate the localized stenotic regions, and the overlaid labels denote the predicted stenosis severity grades.

When mixed axial and sagittal T2-weighted MRI slices are provided as input, the pipeline first performs view classification and subsequently routes each slice to the corresponding view-specific detection and classification models. Central canal stenosis is localized on axial images, while nerve root canal stenosis is identified on sagittal images. For each detected region, a stenosis severity grade—normal, mild, moderate, or severe—is simultaneously assigned, enabling an integrated visualization of anatomical location and clinically relevant severity. This figure illustrates how the proposed pipeline can process raw patient MRI data and present stenosis localization and severity grading in a unified and interpretable manner.

IV. Discussion

In this study, we proposed a two-stage, view-aware AI pipeline for automated localization and severity grading of lumbar spinal stenosis using MRI. By explicitly separating stenotic region localization from severity classification and tailoring each stage to axial and sagittal views, the proposed system reflects the anatomical and clinical characteristics inherent to lumbar spine imaging. This structured detection–classification framework supports consistent automated assessment and may help reduce inter-observer variability and improve diagnostic efficiency in routine clinical practice.

The detection results demonstrated high recall for both central canal and nerve root canal localization, indicating that the YOLOv11-based models effectively captured stenotic regions despite variations in imaging conditions across institutions. However, the relatively lower precision observed in nerve root canal detection suggests that this region remains more challenging than the central canal. This may be attributed to its narrow anatomical space and ambiguous boundaries, which pose challenges for both annotation and model learning.

For severity grading, the DINOv2-based Vision Transformer classifier achieved stable performance across both central canal and nerve root canal assessments. The use of ground-truth–based cropping during training enabled robust learning of severity-related features, while detection-based cropping during inference supported practical applicability in real-world clinical settings. Attention map analysis further confirmed that the model focused on clinically relevant anatomical structures when making predictions.

Although these findings support the potential clinical applicability of the proposed pipeline, further validation is needed to confirm its generalizability across different clinical environments. Independent external validation and direct comparison with clinical experts were not included in this analysis but are currently being conducted as part of an extended study. Future work will include comparisons with prior AI-based methods, further improvement of nerve root canal localization, and analysis of scanner-related differences and annotation variability. Through these additional evaluations, the clinical reliability and applicability of the proposed pipeline are expected to be further clarified.

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